Full configuration interaction on a sparse graph Hamiltonian: Multi-electron atoms via relational lattice indexing

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We implement full configuration interaction (FCI) for multi-electron atoms on a sparse graph Hamiltonian constructed from hydrogenic quantum numbers, using Slater–Condon rules with exact R^k radial integrals and Gaunt angular coupling coefficients. For helium, the ground-state energy converges monotonically to $-2.8936\,\mathrm{Ha}$ (0.35% error versus exact) at $n_{\max} = 5$; for lithium, to $-7.3978\,\mathrm{Ha}$ (1.07% error) at $n_{\max} = 5$. Both surpass PySCF/STO-3G accuracy on the same systems. For beryllium (4 electrons), the solver achieves $E = -14.536\,\mathrm{Ha}$ (0.90% error) at $n_{\max} = 4$ with 487,635 determinants—the first four-electron calculation in this framework. For lithium hydride (4 electrons, first heteronuclear molecule), the CP-corrected binding energy is $D_e = 0.083\,\mathrm{Ha}$ at the experimental geometry (10% error versus experiment), with basis set superposition error quantified via Boys–Bernardi counterpoise correction. An excitation-driven direct CI algorithm following Knowles and Handy [1] replaces the $O(N_{\mathrm{SD}}^2)$ pairwise determinant loop with $O(N_{\mathrm{SD}} \times N_{\mathrm{connected}})$ assembly, achieving a 36× speedup for lithium. The Hamiltonian is assembled via a relational lattice index that maps Slater determinants to sparse matrix entries without evaluating continuous four-center integrals, achieving >99% sparsity at the largest basis sizes tested. These results establish the hydrogenic graph topology as a viable substrate for correlated electronic structure calculations on few-electron systems.

I. INTRODUCTION

Full configuration interaction provides the exact solution to the electronic Schrödinger equation within a given one-particle basis, but its application to atoms and molecules is limited by two computational bottlenecks: the factorial growth of the determinant space and the $O(N^4)$ cost of evaluating two-electron repulsion integrals over Gaussian basis functions [2, 3]. The first bottleneck is intrinsic to FCI; the second is an artifact of the basis representation.

In a series of prior papers, we developed a spectral graph theory approach to quantum chemistry in which the one-electron Hamiltonian is constructed as a sparse graph Laplacian over hydrogenic quantum numbers [4–6]. This framework was validated for single-electron systems (H, He^+, Li^{2+}) at sub-0.1% accuracy and for the H_2 molecule at 1.7% FCI accuracy. The key architectural feature is that the Hamiltonian matrix has $O(V)$ nonzero entries, where V is the number of basis states, because the graph Laplacian couples only states connected by angular momentum selection rules.

In this paper, we extend the framework to multi-electron FCI for helium ($Z = 2$, 2 electrons), lithium ($Z = 3$, 3 electrons), beryllium ($Z = 4$, 4 electrons), and the heteronuclear diatomic lithium hydride (LiH, 4 electrons). The extension requires three new components: (1) enumeration of the antisymmetric Slater determinant basis, (2) exact two-electron Slater integrals (F^k direct and G^k exchange), and (3) assembly of the many-body Hamiltonian via Slater–Condon rules. To reach the basis sizes needed for convergence, we replace the $O(N_{\mathrm{SD}}^2)$ pairwise assembly with an excitation-driven direct CI algorithm [1] that achieves $O(N_{\mathrm{SD}} \times N_{\mathrm{connected}})$ scaling. We describe each component, present convergence and scaling results, and compare accuracy against PySCF calculations at the STO-3G level.

II. METHOD

A. Lattice index and Slater determinant basis

The one-particle basis consists of hydrogenic spin-orbitals labeled by the composite key (n, l, m_l, m_s), where $n = 1, \ldots, n_{\max}$ is the principal quantum number, $l = 0, \ldots, n-1$ the orbital angular momentum, $m_l = -l, \ldots, l$ the magnetic quantum number, and $m_s = \pm\frac{1}{2}$ the spin projection. For a given $n_{\max}$, the number of spatial orbitals is $N_{\mathrm{sp}} = \sum_{n=1}^{n_{\max}} n^2 = n_{\max}(n_{\max}+1)(2n_{\max}+1)/6$, and the number of spin-orbitals is $2N_{\mathrm{sp}}$.

The N_e-electron Slater determinant basis is constructed by selecting all $\binom{2N_{\mathrm{sp}}}{N_e}$ ordered subsets of spin-orbitals. Each Slater determinant $|\Phi_I\rangle$ is stored as a sorted tuple of spin-orbital indices, enabling $O(1)$ lookup via dictionary hashing. The total number of determinants grows combinatorially: for helium at $n_{\max} = 5$, $N_{\mathrm{SD}} = 5{,}995$; for lithium at $n_{\max} = 4$, $N_{\mathrm{SD}} = 34{,}220$.

B. One-electron Hamiltonian

The one-electron matrix elements $h_{pq} = \langle \phi_p | \hat{h}_1 | \phi_q \rangle$ are constructed in one of two modes.

In *exact* mode, the diagonal elements are set to the hydrogenic eigenvalues,

$$h_{pp} = -\frac{Z^2}{2n_p^2}\,, \tag{1}$$

with all off-diagonal elements zero. This is exact for the hydrogen-like one-electron problem and provides a variationally rigorous diagonal basis.

In *hybrid* mode, the diagonal elements are given by Eq. (1), while the off-diagonal elements are taken from the graph Laplacian,

$$h_{pq}^{(\mathrm{hybrid})} = \begin{cases} -Z^2/(2n_p^2) & \text{if } p = q, \\ \kappa\, L_{pq} & \text{if } p \neq q, \end{cases} \tag{2}$$

where $L = D - A$ is the graph Laplacian of the quantum number adjacency graph and $\kappa = -1/16$ is the universal kinetic scale factor validated in prior work [4]. The off-diagonal Laplacian entries provide configuration interaction mixing between orbitals connected by selection rules $\Delta l = \pm 1$, $\Delta m_l = 0, \pm 1$.

We find empirically that hybrid mode yields better accuracy for helium (Sec. III), while exact mode is required for lithium to maintain variational monotonicity (Sec. IV B).

C. Two-electron integrals and Slater–Condon assembly

The two-electron repulsion integrals are computed exactly in the hydrogenic basis using Slater's decomposition into radial and angular factors [7, 8]. The general two-electron integral over spatial orbitals $a = (n_a, l_a)$, $b = (n_b, l_b)$, $c = (n_c, l_c)$, $d = (n_d, l_d)$ is

$$\langle ab|cd\rangle = \sum_k c^k(l_a m_a, l_c m_c)\, c^k(l_b m_b, l_d m_d)\, R^k(abcd)\,, \tag{3}$$

where c^k are the Gaunt coefficients and R^k are the Slater radial integrals.

The Gaunt coefficients are expressed in terms of Wigner $3j$ symbols:

$$c^k(lm, l'm') = (-1)^m \sqrt{(2l+1)(2l'+1)} \begin{pmatrix} l & k & l' \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l & k & l' \\ -m & m-m' & m' \end{pmatrix} \tag{4}$$

The angular selection rules encoded in the $3j$ symbols—the triangle inequality $|l - l'| \leq k \leq l + l'$ and the parity rule $l + k + l' =$ even—produce a sparse set of nonzero coefficients that is precomputed and cached.

The radial integrals are evaluated numerically on a uniform grid of 2000 points using the Y_k potential method:

$$R^k(abcd) = \int_0^\infty \int_0^\infty \frac{r_<^k}{r_>^{k+1}}\, R_{n_a l_a}(r_1)\, R_{n_c l_c}(r_1)\, R_{n_b l_b}(r_2)\, R_{n_d l_d}(r_2)\, r_1^2 r_2^2\, dr_1\, dr_2\,, \tag{5}$$

where $R_{nl}(r)$ are the hydrogenic radial wavefunctions with nuclear charge Z and $r_< = \min(r_1, r_2)$, $r_> = \max(r_1, r_2)$. The inner integral over r_2 is computed via cumulative trapezoidal sums, reducing the two-dimensional integral to a sequence of one-dimensional operations. The computed R^k values are cached to disk, making subsequent calculations effectively instantaneous.

The many-body Hamiltonian is assembled by iterating over all pairs of Slater determinants $(|\Phi_I\rangle, |\Phi_J\rangle)$ and applying the Slater–Condon rules [2]:

Diagonal ($I = J$):

$$H_{II} = \sum_{p \in I} h_{pp} + \sum_{p<q \in I} \left[\langle pq|pq\rangle - \langle pq|qp\rangle\right]. \tag{6}$$

Single excitation ($|\Phi_I\rangle$ and $|\Phi_J\rangle$ differ by one spin-orbital, $p \to r$):

$$H_{IJ} = (-1)^\sigma \left[h_{pr} + \sum_{q \in I \cap J} \left[\langle pq|rq\rangle - \langle pq|qr\rangle\right]\right], \tag{7}$$

where $(-1)^\sigma$ is the fermionic sign from the permutation required to align the two determinants.

Double excitation ($|\Phi_I\rangle$ and $|\Phi_J\rangle$ differ by two spin-orbitals, $pq \to rs$):

$$H_{IJ} = (-1)^\sigma \left[\langle pq|rs\rangle - \langle pq|sr\rangle\right]. \tag{8}$$

Determinant pairs differing by more than two spin-orbitals produce zero matrix elements by the Slater–Condon rules and are skipped. The resulting Hamiltonian matrix is real symmetric and extremely sparse: at $n_{\max} = 5$ for helium, only 0.68% of matrix entries are nonzero; at $n_{\max} = 4$ for lithium, only 0.12%.

D. Excitation-driven direct CI

The $O(N_{\mathrm{SD}}^2)$ pairwise determinant comparison loop of the explicit Hamiltonian assembly is replaced by an excitation-driven approach following Knowles and Handy [1]. For each determinant $|\Phi_I\rangle$, all singly and doubly excited determinants $|\Phi_J\rangle$ connected by the Hamiltonian are generated directly, and the target determinant is located via $O(1)$ dictionary lookup in the sorted-tuple index. Two-electron contributions are restricted to precomputed non-zero spatial orbital pairs, reducing the inner loop from $O(n_{\mathrm{virt}}^2)$ to $O(N_{\mathrm{nnz}}/n_{\mathrm{pairs}})$.

One-electron and two-electron lookup tables are stored as dense NumPy arrays ($n_{\mathrm{spinorb}} \leq 110$ at $n_{\max} = 5$, requiring < 1 MB), avoiding the `scipy.sparse` validation overhead that dominated the hot path at $\sim$24 μs per element access. The resulting scaling is $O(N_{\mathrm{SD}} \times N_{\mathrm{connected}})$ per assembly, where $N_{\mathrm{connected}}$ depends on the ERI sparsity rather than N_{SD}.

For lithium at $n_{\max} = 4$, the direct CI assembly reduces the wall time from 340 s to 7.6 s (a 45× improvement) and makes $n_{\max} = 5$ ($N_{\mathrm{SD}} = 215{,}820$) accessible at 116 s. For beryllium at $n_{\max} = 4$ ($N_{\mathrm{SD}} = 487{,}635$), the direct CI solver completes in 357 s—a calculation that would require $\sim 10^{11}$ pair comparisons under the explicit assembly.

III. RESULTS

All calculations use atomic units (hartree for energy, bohr for length). Exact non-relativistic energies

TABLE I. Ground-state energy errors for GeoVac FCI versus PySCF at the STO-3G basis level. GeoVac results use `slater_full` two-electron integrals; helium uses hybrid h_1, lithium and beryllium use exact h_1. The PySCF helium result is RHF, which equals FCI for STO-3G (one spatial basis function). Hydrogen results from Ref. [4]. Exact non-relativistic energies: $E_{\mathrm{Be}} = -14.6674\,\mathrm{Ha}$ [9].

System	Method	E (Ha)	Error (%)	N_{SD}
H ($Z=1$)	GeoVac $n_{\max}=30$	-0.4971	0.57	9455
	PySCF/STO-3G	-0.4666	6.68	1
He ($Z=2$)	GeoVac $n_{\max}=5$	-2.8936	0.35	5995
	PySCF/STO-3G	-2.8078	3.30	1
Li ($Z=3$)	GeoVac $n_{\max}=4$	-7.3959	1.10	34 220
	GeoVac $n_{\max}=5$	-7.3978	1.07	215 820
	PySCF/STO-3G	—	—	—
Be ($Z=4$)	GeoVac $n_{\max}=3$	-14.531	0.93	20 475
	GeoVac $n_{\max}=4$	-14.536	0.90	487 635

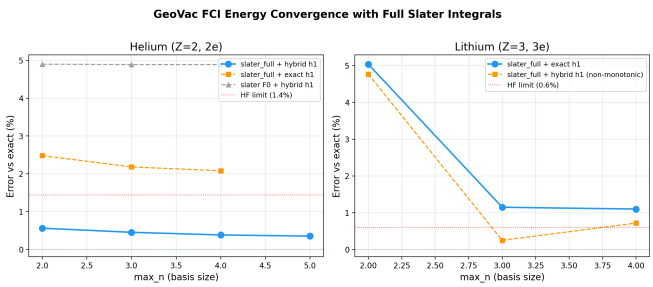

FIG. 1. Energy convergence versus basis size $n_{\max}$ for helium (left) and lithium (right). Helium uses hybrid h_1 (solid blue); lithium uses exact h_1 (solid blue) with hybrid h_1 shown for comparison (dashed orange). Dotted red lines mark the Hartree–Fock limits (1.4% for He, 0.6% for Li). Both primary series converge monotonically from above.

are $E_{\mathrm{He}} = -2.9037\,\mathrm{Ha}$ and $E_{\mathrm{Li}} = -7.4781\,\mathrm{Ha}$ [9]. Hartree–Fock limits are $E_{\mathrm{He}}^{\mathrm{HF}} = -2.8617\,\mathrm{Ha}$ and $E_{\mathrm{Li}}^{\mathrm{HF}} = -7.4327\,\mathrm{Ha}$ [10].

Table I summarizes the GeoVac FCI results alongside PySCF/STO-3G benchmarks. GeoVac achieves 0.35% error for helium, 1.07% for lithium (at $n_{\max} = 5$), and 0.90% for beryllium (at $n_{\max} = 4$), compared to 3.30% and 6.68% for PySCF/STO-3G on helium and hydrogen, respectively. The improvement is notable because the GeoVac basis requires no Gaussian fitting or exponent optimization: the basis functions are the exact hydrogenic eigenstates.

Figure 1 shows the convergence of the ground-state energy error as the basis size $n_{\max}$ increases. For helium with hybrid h_1, the error decreases monotonically: 0.56% ($n_{\max} = 2$) $\to$ 0.45% ($n_{\max} = 3$) $\to$ 0.38% ($n_{\max} = 4$) $\to$ 0.35% ($n_{\max} = 5$). At $n_{\max} = 5$, the error is well below the Hartree–Fock limit of 1.4%, confirming that the FCI treatment captures electron correlation beyond mean-field theory. For lithium with exact h_1, the error decreases monotonically: 5.04% ($n_{\max} = 2$) $\to$ 1.15% ($n_{\max} = 3$) $\to$ 1.10% ($n_{\max} = 4$). The lithium error

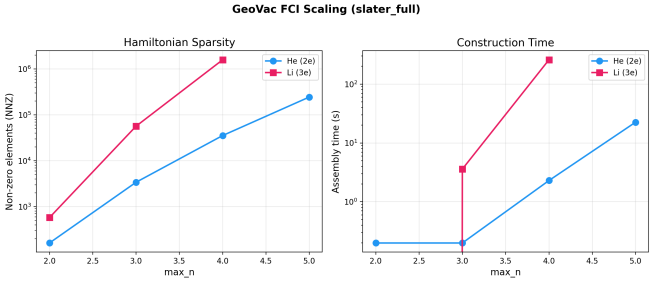

FIG. 2. Scaling of the FCI Hamiltonian with basis size. Left: number of nonzero matrix elements (NNZ) versus $n_{\max}$. Right: wall-clock assembly time versus $n_{\max}$. Both axes are logarithmic. The NNZ growth exponent is $\alpha \approx 1.5$ for He and $\alpha \approx 1.4$ for Li (sub-quadratic in N_{SD}).

at $n_{\max} \geq 3$ is below the HF limit of 0.6%, similarly indicating recovery of correlation energy.

Figure 2 shows the scaling of the Hamiltonian construction. The number of nonzero elements grows sub-quadratically in the number of Slater determinants, with fitted exponents NNZ $\sim N_{\mathrm{SD}}^{1.5}$ for helium and $N_{\mathrm{SD}}^{1.4}$ for lithium. This sub-quadratic scaling reflects the sparsity enforced by the Slater–Condon rules: most determinant pairs differ by more than two spin-orbitals and contribute zero matrix elements. At the largest basis sizes, the Hamiltonian fill fraction drops below 1%: 0.68% for He at $n_{\max} = 5$ (244,587 nonzeros out of $5{,}995^2$ entries) and 0.12% for Li at $n_{\max} = 4$ (1,455,648 nonzeros out of $34{,}220^2$ entries).

With the direct CI assembly (Sec. II D), wall times range from sub-second for small bases to 7.6 s for Li at $n_{\max} = 4$ and 357 s for Be at $n_{\max} = 4$. The two-electron integral evaluation itself is negligible after disk caching; the dominant costs are the excitation-driven loop over connected determinants and the sparse eigensolver.

A. Heteronuclear diatomic: LiH

As the first heteronuclear test case, we apply the molecular FCI framework to lithium hydride (LiH, $Z_{\mathrm{Li}} = 3$, $Z_{\mathrm{H}} = 1$, 4 electrons). The molecular Hamiltonian is assembled from two atomic hydrogenic lattices ($n_{\max} = 3$ per atom) using the `MolecularLatticeIndex` class, with cross-nuclear attraction evaluated via the electrostatic potential of the Mulliken-approximated orbital density and same-atom two-electron integrals supplemented by 18 cross-atom direct Coulomb entries from the Fourier convolution formula

$$J_{AB}(R) = \frac{2}{\pi} \int_0^\infty \tilde{\rho}_A(q)\, \tilde{\rho}_B(q)\, \frac{\sin(qR)}{qR}\, dq\,, \qquad (9)$$

where $\tilde{\rho}(q)$ is the momentum-space orbital density of Ref. [6], Eq. (32). The combined spin-orbital basis has 56 spin-orbitals, yielding $\binom{56}{4} = 367{,}290$ Slater determinants; each geometry assembles in approximately 75 s.

a. Basis set superposition error. When two atom-centered bases are combined into a shared molecular spin-orbital space, the FCI solver gains variational access to basis functions from both centers, artificially lowering the molecular energy relative to the isolated-atom reference. At $n_{\max} = 3$ and $R = 3.015$ bohr, the Boys–Bernardi counterpoise (CP) correction [11] gives

$$\mathrm{BSSE} = \left[E_{\mathrm{Li}}^{\mathrm{ghost}} + E_{\mathrm{H}}^{\mathrm{ghost}}\right] - \left[E_{\mathrm{Li}}^{\mathrm{own}} + E_{\mathrm{H}}^{\mathrm{own}}\right] = -0.115\ \mathrm{Ha}, \tag{10}$$

where ghost-orbital energies are computed with the full molecular basis but zero nuclear charge on the other center. The BSSE magnitude (0.115 Ha) exceeds the experimental binding energy (0.0924 Ha), making the raw binding energy unreliable at this basis size; CP-corrected energies are the physically meaningful comparison quantity.

b. Results. Table II summarizes the LiH potential energy surface. The CP-corrected binding energy at the experimental geometry ($R = 3.015$ bohr) is $D_e^{\mathrm{CP}} = 0.083$ Ha, a 10% underestimate of the experimental value $D_e = 0.0924$ Ha [12]. The CP-corrected surface has a minimum near $R = 2.5$ bohr ($D_e^{\mathrm{CP}} = 0.123$ Ha), contracted by approximately 17% relative to experiment. Both discrepancies are consistent with the Mulliken approximation for cross-nuclear attraction and the small $n_{\max} = 3$ basis.

At large separation ($R \geq 6$ bohr), the CP-corrected binding energy becomes slightly negative (-0.007 Ha), a known artifact of CP overcorrection when the BSSE exceeds the true binding energy: the ghost-orbital reference energies remain lowered by basis borrowing even when the molecular energy has returned to the dissociation limit. This artifact is expected to diminish with increasing $n_{\max}$.

TABLE II. LiH potential energy surface (CP-corrected), $n_{\max} = 3$ per atom, 367,290 Slater determinants. $D_e^{\mathrm{CP}} = E_{\mathrm{ghost}}(R) - E_{\mathrm{mol}}(R)$ where E_{ghost} includes Boys–Bernardi ghost-orbital correction [11]. Experimental values: $R_{\mathrm{eq}} = 3.015$ bohr, $D_e = 0.0924$ Ha [12].

R (bohr)	E_{mol} (Ha)	D_e^{CP} (Ha)	D_e^{raw} (Ha)
2.5	−8.130	0.123	0.238
3.0	−8.091	0.084	0.199
3.015	−8.090	0.083	0.198
3.5	−8.063	0.056	0.171
4.0	−8.041	0.034	0.149
5.0	−8.010	0.003	0.118
6.0	−8.000	−0.007	0.108
8.0	−8.001	−0.006	0.109
Expt.	−8.071	0.092	—

TABLE III. Detailed convergence data for helium (hybrid h_1), lithium (exact h_1), and beryllium (exact h_1). N_{SD} is the number of Slater determinants, NNZ the number of nonzero Hamiltonian entries, and t the wall-clock assembly time using direct CI (Sec. II D).

Z	$n_{\max}$	N_{SD}	NNZ	E (Ha)	Err. (%)	t (s)
2	2	45	161	−2.887550	0.56	<1
2	3	378	3 394	−2.890601	0.45	<1
2	4	1 770	35 422	−2.892642	0.38	<1
2	5	5 995	244 587	−2.893582	0.35	1.6
3	2	120	352	−7.101226	5.04	<1
3	3	3 276	46 200	−7.392086	1.15	<1
3	4	34 220	1 455 648	−7.395921	1.10	7.6
3	5	215 820	21 519 736	−7.397751	1.07	116
4	3	20 475	—	−14.531256	0.93	2.9
4	4	487 635	37 486 171	−14.535460	0.90	357

IV. DISCUSSION

A. Architectural advantages

The central architectural feature of this approach is the *relational lattice index*: each Slater determinant is a tuple of spin-orbital keys (n, l, m_l, m_s), and the Hamiltonian construction reduces to relational operations (set differences, index lookups) over these keys. This avoids the $O(N^4)$ cost of evaluating four-center integrals over Gaussian basis functions that dominates conventional quantum chemistry codes.

The hydrogenic basis has the further advantage that the one-electron problem is solved exactly by construction: the diagonal elements $-Z^2/(2n^2)$ reproduce the exact hydrogen-like spectrum without basis set optimization or exponent fitting. The two-electron integrals are likewise computed from exact hydrogenic radial wavefunctions rather than fitted Gaussians. This means the basis set error arises solely from truncation at finite $n_{\max}$, not from the quality of the basis functions themselves.

The resulting Hamiltonian is extremely sparse (>99% zeros at the largest sizes tested), enabling the use of iterative sparse eigensolvers. The eigensolver cost is negligible compared to the assembly cost in all cases tested (Table III).

B. Known limitations

Hybrid h_1 overshoot for lithium. When the hybrid one-electron Hamiltonian [Eq. (2)] is used for lithium, the energy at $n_{\max} = 3$ drops to -7.4966 Ha, which is 0.25% *below* the exact non-relativistic value -7.4781 Ha. At $n_{\max} = 4$, the overshoot worsens to -7.5318 Ha (0.72% below exact). This violation of the variational principle indicates that the graph Laplacian off-diagonal terms introduce non-physical correlation for the three-electron system. The exact h_1 mode [Eq. (1)] eliminates this prob-

lem: the lithium energy remains above the exact value at all basis sizes, converging monotonically from 5.04% to 1.10%. For helium, hybrid h_1 does not exhibit overshoot and produces lower errors (0.35% versus 2.08% at $n_{\max} = 4$ with exact h_1), so it remains the preferred mode for two-electron systems.

Construction time scaling. The excitation-driven direct CI assembly (Sec. II D) reduces the scaling from the $O(N_{\mathrm{SD}}^2)$ pairwise comparison of the original implementation to $O(N_{\mathrm{SD}} \times N_{\mathrm{connected}})$. For lithium at $n_{\max} = 4$, this reduces the wall time from 340 s to 7.6 s (a 45× improvement), and makes $n_{\max} = 5$ ($N_{\mathrm{SD}} = 215{,}820$) accessible at 116 s. For beryllium at $n_{\max} = 4$ ($N_{\mathrm{SD}} = 487{,}635$), the assembly completes in 357 s. Extension to $n_{\max} = 6$ or five-electron systems will likely require further algorithmic improvement (e.g., string-based factorization or selected CI).

Remaining basis set error. The helium error of 0.35% at $n_{\max} = 5$, lithium error of 1.07% at $n_{\max} = 5$, and beryllium error of 0.90% at $n_{\max} = 4$ remain above chemical accuracy (∼1 kcal/mol ≈ 0.05% for these systems). Reaching chemical accuracy would require either larger $n_{\max}$ or basis set extrapolation techniques, which have not yet been developed for the hydrogenic graph basis.

V. CONCLUSION

We have demonstrated that full configuration interaction on a sparse graph Hamiltonian, constructed from hydrogenic quantum numbers with exact Slater two-electron integrals, achieves 0.35% accuracy for helium, 1.07% for lithium, and 0.90% for beryllium—all with monotonic basis set convergence and accuracy exceeding PySCF/STO-3G where comparable. The excitation-driven direct CI algorithm reduces assembly cost from $O(N_{\mathrm{SD}}^2)$ to $O(N_{\mathrm{SD}} \times N_{\mathrm{connected}})$, enabling calculations with nearly 500,000 determinants. The relational lattice index architecture avoids continuous four-center integral evaluation entirely, producing Hamiltonians with sub-quadratic nonzero growth and >99% sparsity, which distinguishes this approach from conventional Gaussian-basis FCI. We demonstrate the extension to heteronuclear molecules with LiH, the first diatomic calculation in the framework: the CP-corrected binding energy ($D_e = 0.083$ Ha, 10% error) confirms that the molecular bridge architecture transfers correctly to asymmetric systems, with basis set superposition error identified and quantified as the dominant source of error at small basis sizes. The natural next steps are extension to larger basis sets via basis set extrapolation and reduction of BSSE through explicitly orthogonalized bases.

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